

Data-Driven Safety Prioritization for NYC Uncontrolled Intersections

Maggie Boles, Adina Kaplan, Logan McCorkle, & Chana Ochs

Bellevue University

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Professor Fadi Alsaleem

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Data-Driven Safety Prioritization for NYC Uncontrolled Intersections

New York City's Vision Zero initiative is built on the belief that traffic fatalities are preventable through data-driven policy. While major crossings are heavily regulated, thousands of "uncontrolled" intersections remain a safety challenge. This project utilizes a predictive model to identify high-risk uncontrolled intersections, providing the Department of Transportation (DOT) with a proactive roadmap to intervene before accidents occur.

Business Problem and Hypothesis

The NYC DOT manages a massive network with finite resources. Currently, infrastructure improvements are largely reactive; the city often installs signals only after a location has established a history of injuries.

We hypothesize that an intersection's risk level is a product of its structural and geographic signature. By synthesizing geographic variables with localized data patterns, our model can identify "latent risk", the high probability of a future crash, even at intersections that have not yet seen a major incident.

Methods and Analysis

Our analysis utilized four primary datasets, which were cleaned and standardized for modeling (see Appendix A):

- **NYC Motor Vehicle Collisions** includes historical crash data filtered from 2022 to March 2026.
- **Traffic Signal and All-Way Stop Study Requests** contains community requests for new traffic controls.
- **Traffic Signal Condition** includes 311-style reports regarding maintenance and signal issues.

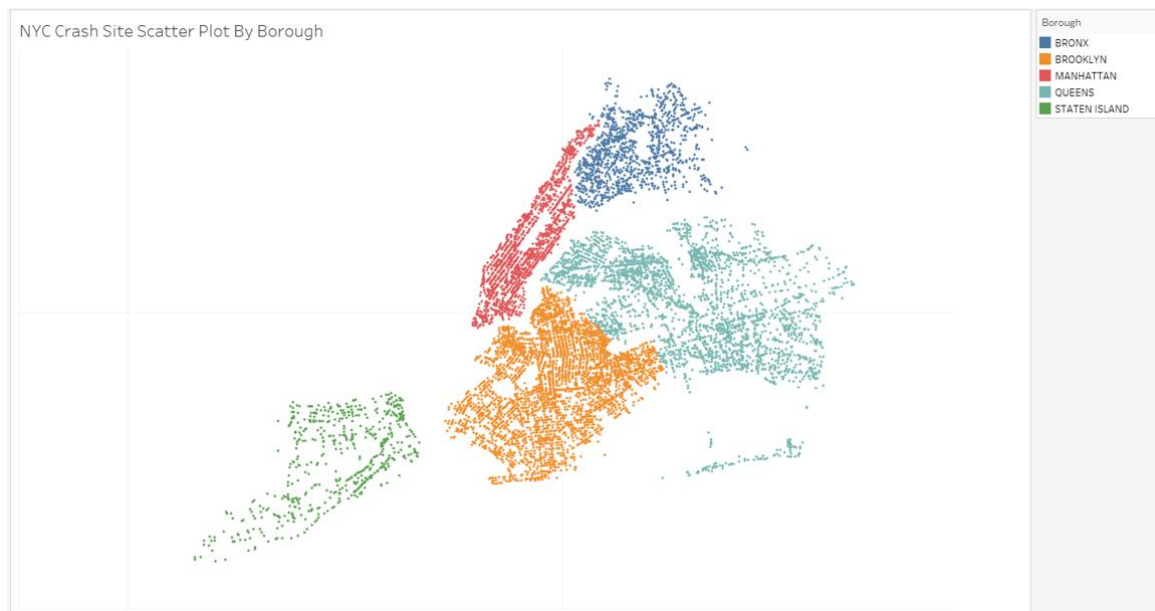
- **NYS DOT Maintained Traffic Signals** provides a comprehensive inventory of existing signal infrastructure.

The integration of these disparate sources required a standardized spatial framework.

Figure 1 shows the geographic distribution of these combined records, highlighting the density of incident reports across the five boroughs.

Figure 1

Geographic Distribution of Integrated Crash Data (2022–2026)

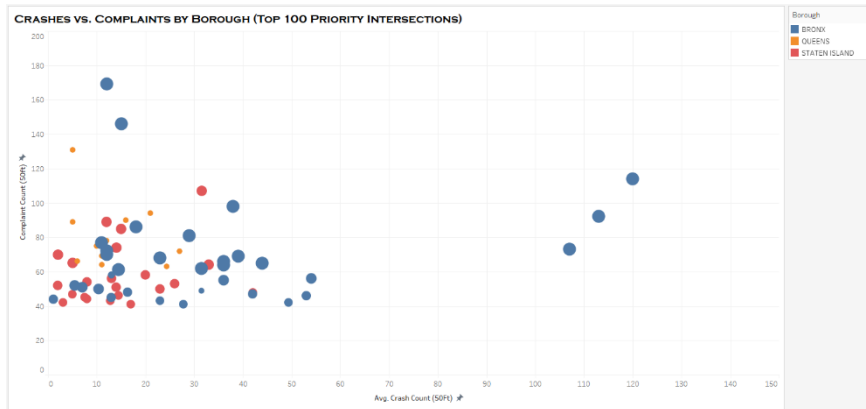


Model Selection

We used a Random Forest Classifier with 200 decision trees (see Appendix B). This method is ideal for urban safety because it identifies the specific mathematical signature of a dangerous intersection by examining how variables such as geographic coordinates (X and Y State Plane) and borough characteristics overlap (see Appendix C). Figure 2 illustrates one such signature, showing the statistical correlation between community complaint volume and historical crash frequency.

Figure 2

Correlation Signature: Community Complaint Volume vs. Average Crash Frequency



Evaluation and Recall

We prioritized recall over simple accuracy, achieving a score of 0.97 (see Appendix D). This ensures that our priority list captures 97% of historically dangerous sites.

Results

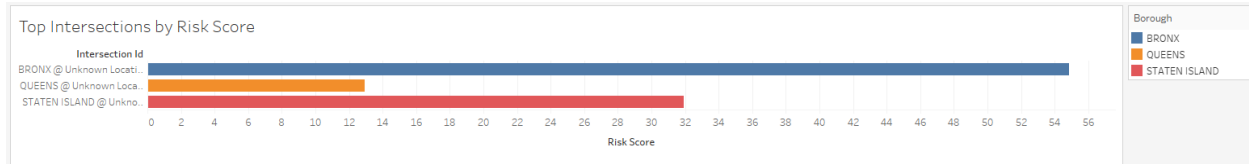
The model identified 100 high-priority intersections for immediate audit. Data analysis reveals a significant geographic concentration of risk, with the majority of flagged locations situated in the Bronx. The distribution across boroughs is as follows:

- Bronx: 55 intersections (55%)
- Staten Island: 32 intersections (32%)
- Queens: 13 intersections (13%)

As illustrated in Figure 3, the disparity in risk scores highlights the specific boroughs where infrastructure intervention would yield the highest safety ROI.

Figure 3

Distribution of High-Priority Intersections and Total Risk Scores by Borough



Recommendations and Ethical Considerations

Recommendations

To address the identified risks, the study recommends the following operational steps:

- Dispatch field teams to audit the high-risk locations for structural issues.
- Fast-track low-cost interventions for any site with a risk score above 0.95.

Ethical Considerations

Deploying this predictive model requires a rigorous evaluation of ethical considerations to ensure the equitable distribution of safety resources:

- **Impact:** Proactive safety improvements save lives.
- **Assumptions and Bias:** We assume reporting is uniform; however, neighborhoods with higher digital literacy may report issues more often, potentially biasing the model.
- **Privacy:** All data used was sourced from public records and is anonymized.
- **Misinterpretation:** Risk Scores are probabilities; they support, rather than replace, engineering judgment.
- **Responsibility:** The team must audit the model annually to ensure that safety improvements are distributed equitably across all communities.

Conclusion

The transition from reactive maintenance to proactive risk mitigation is facilitated by predictive modeling. This analysis demonstrates that by isolating specific risk signatures through the synthesis of spatial and community data, the city can more effectively align its resources with Vision Zero objectives.

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Appendix A

Data Preparation and Cleaning

The modeling process involved standardizing crash dates and times and filtering for recent records (2022–2026). To ensure model integrity, rows with missing critical latitude/longitude data were removed, and the dataset was balanced to achieve equal distribution of risk levels. This prevents the model from becoming biased toward "low-risk" outcomes simply because they are more common in the raw data.

Appendix B

Technical Methodology: Random Forest Classifier

The team utilized a Random Forest Classifier because it identifies complex "if-then" patterns between coordinates and infrastructure data that simpler models often miss. The algorithm constructs a multitude of decision trees during training. For classification, the output is the class selected by the majority of the trees; this "ensemble" approach prevents overfitting and allows the model to identify stable "risk signatures" across the city's vast intersection network.

Appendix C

Spatial Data: NY State Plane (X, Y) Coordinates

To ensure precision across administrative "Unknowns," we utilized the EPSG:2263 (New York Long Island) State Plane coordinate system. Unlike standard latitude/longitude, this system uses a flat grid measured in feet based on a Transverse Mercator projection. This enables high-accuracy distance calculations and spatial joins across the five boroughs, which are essential for identifying the proximity of crashes to specific uncontrolled intersections.

Appendix D

Evaluation Metrics

In safety modeling, missing a dangerous intersection involves a much higher cost than over-inspecting a safe one. While accuracy measures overall correctness, recall measures the model's ability to find all actual high-risk cases. By prioritizing a high recall score, the team ensures that 97% of historically dangerous locations are successfully flagged for audit, minimizing false negatives that could lead to preventable accidents.

Figure 4

Confusion Matrix: Performance Metrics for Random Forest Classifier

Optimized Accuracy: 0.8842

Final Classification Report:

	precision	recall	f1-score	support
0	0.96	0.79	0.87	2615
1	0.84	0.97	0.90	2836
accuracy			0.88	5451
macro avg	0.90	0.88	0.88	5451
weighted avg	0.89	0.88	0.88	5451